

A Kummer Distribution Criterion for Erdős–Graham Problem #203

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Abstract

We formulate a precise Kummer distribution criterion sufficient for the negative resolution of Erdős–Graham Problem #203. The criterion, called prime Hecke non-concentration (PHNC), asserts that fixed-radical Kummer characters do not concentrate almost entirely at the identity among primes $q \leq Q$, $q \equiv 1 \pmod{\ell}$, uniformly for $\ell \leq (\log Q)^B$. This note separates the exact algebraic reductions from the currently open analytic high- ℓ seam. It is a conditional criterion, not an unconditional solution of #203.

1 The problem

For $(m, 6) = 1$, define

$$A(m) = \{(k, \ell) \in \mathbb{Z}_{\geq 0}^2 : m2^k3^\ell + 1 \in \mathbb{P}\}.$$

The desired negative resolution of Erdős–Graham Problem #203 is

$$\forall m, (m, 6) = 1, \quad |A(m)| = \infty.$$

2 Kummer vectors

Let ℓ be prime and $q \equiv 1 \pmod{\ell}$. Choose a primitive root $g_q \pmod{q}$, and write

$$2 = g_q^{u_q}, \quad 3 = g_q^{v_q}.$$

Define

$$x_\ell(q) = (u_q, v_q) \in \mathbb{F}_\ell^2.$$

For $(r, s) \in \mathbb{F}_\ell^2$, define the Kummer character condition

$$ru_q + sv_q \equiv 0 \pmod{\ell},$$

equivalently

$$2^r 3^s \in (\mathbb{F}_q^*)^\ell.$$

3 Prime Hecke non-concentration

Hypothesis 1 (PHNC). *For every fixed $B > 0$, there exists $C = C(B) > 0$ such that, for every prime*

$$\ell \leq (\log Q)^B$$

and every nonzero $(r, s) \in \mathbb{F}_\ell^2$,

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, 2^r 3^s \in (\mathbb{F}_q^*)^\ell\} \leq (1 - \ell^{-C})\pi(Q; \ell, 1)$$

for all sufficiently large Q .

Remark 1. *PHNC is a non-concentration statement, not an equidistribution statement. Equidistribution would predict relative density $1/\ell$. PHNC only forbids relative density $1 - o(\ell^{-C})$.*

4 Kummer-BV criterion

A stronger but more analytic form is a Kummer Bombieri–Vinogradov type variance theorem.

Hypothesis 2 (Kummer-BV). *For every $A, B > 0$, a suitable prime weight ω_q , and all Q ,*

$$\sum_{\ell \leq (\log Q)^B} \ell^{\alpha-2} \sum_{(r,s) \neq (0,0)} \left| \sum_{\substack{q \leq Q \\ q \equiv 1 \pmod{\ell}}} \omega_q e_\ell(r \log_q 2 + s \log_q 3) \right|^2 \ll_A (\log Q)^{-A}.$$

Remark 2. *The precise weight and exponent α must be matched to the sieve application. This draft records the structural role of the criterion.*

5 PHNC to smoothing

Hypothesis 3 (Random-walk smoothing bridge). *PHNC implies a polynomial spectral gap for the Kummer-vector measures $x_\ell(q)$, and after sufficiently many convolutions yields collision probability*

$$\ell^{-2} + O_A((\log Q)^{-A})$$

uniformly in the required ℓ -range.

Remark 3. *This finite-field smoothing bridge is an explicit auxiliary lemma to prove. One possible formulation is: if a measure on $\mathbb{F}_\ell^2$ assigns at most $1 - \delta$ mass to every hyperplane, then its nontrivial Fourier coefficients are bounded away from 1 by a polynomial in δ , possibly under an anti-atom condition.*

6 Conditional closure theorem

Theorem 1 (Conditional Erdős–Graham #203 criterion). *Assume:*

1. *PHNC, Hypothesis 1;*
2. *the random-walk smoothing bridge, Hypothesis 3;*

3. a rank-two Kummer large-sieve/Kummer-BV estimate strong enough for the BVK sieve;
4. the BVK-to-NEG-203 sieve implication.

Then for every $m \in \mathbb{Z}_{>0}$ with $(m, 6) = 1$,

$$|A(m)| = \infty.$$

Proof architecture. The implication chain is

PHNC $\Rightarrow$ KRWS $\Rightarrow$ Kummer residue collision control $\Rightarrow$ rank-two Hecke large sieve $\Rightarrow$ Kummer-BV $\Rightarrow$ BVK sieve

The first arrow is Hypothesis 3. The Kummer residue collision control converts smoothed vector distribution into pair-collision bounds for the projective slopes. The rank-two Hecke large sieve converts those collision bounds into the Kummer-BV variance estimate. The BVK sieve then supplies infinitely many primes of the form $m2^k3^\ell + 1$. Each arrow is a dependency. A complete paper must formalize or cite every arrow separately; this draft is an architecture document until that is done. $\square$

7 Low- ℓ range

For a fixed nonzero (r, s) , let

$$K_\ell = \mathbb{Q}(\zeta_\ell), \quad L_{\ell,r,s} = K_\ell((2^r 3^s)^{1/\ell}).$$

By valuation at primes over 2 and 3, $2^r 3^s$ is not an ℓ -th power in K_ℓ when $(r, s) \neq 0$. Thus

$$[L_{\ell,r,s} : K_\ell] = \ell.$$

Effective Chebotarev/log-free Hecke estimates give PHNC in ranges where the field/conductor parameter is sufficiently small relative to Q , heuristically

$$\ell \log \ell \ll \log Q.$$

A safe statement should cite the exact effective theorem and its field-invariant dependence.

8 The high- ℓ seam

The open seam is

$$(\log Q)^{1-\varepsilon} < \ell \leq (\log Q)^B.$$

Fixed-field ineffective Chebotarev is not enough to prove PHNC uniformly in this growing family. The required theorem is the following.

Problem 1 (Uniform fixed-radical Kummer non-concentration). *For every $B > 0$, prove that there exists $C = C(B) > 0$ such that PHNC holds for all*

$$\ell \leq (\log Q)^B$$

and all nonzero $(r, s) \in \mathbb{F}_\ell^2$.

9 GRH conditional variant

Hypothesis 4 (Kummer GRH). *Assume GRH for the Hecke/Artin L-functions attached to the fields*

$$\mathbb{Q}(\zeta_\ell, (2^r 3^s)^{1/\ell})$$

uniformly for $\ell \leq (\log Q)^B$ and nonzero (r, s) .

Corollary 1 (GRH-conditional route). *Under a sufficiently uniform Kummer-GRH Chebotarev estimate, the PHNC criterion holds, and hence Theorem 1 applies.*

Proof. Uniform GRH Chebotarev gives strong splitting estimates in each cyclic Kummer layer. These imply non-concentration at the identity value. The conditional closure theorem then applies. $\square$

10 Expert-facing formulation

The expert-facing question extracted from this paper is:

Let

$$K_\ell = \mathbb{Q}(\zeta_\ell), \quad \chi_{r,s} = \left(\frac{2}{\cdot}\right)_\ell^r \left(\frac{3}{\cdot}\right)_\ell^s$$

for nonzero $(r, s) \in \mathbb{F}_\ell^2$. Does one have a uniform polynomial non-concentration estimate

$$\#\{q \leq Q : q \equiv 1 \pmod{\ell}, \chi_{r,s}(q) = 1\} \leq (1 - \ell^{-C(B)})\pi(Q; \ell, 1)$$

for all $\ell \leq (\log Q)^B$?

This is the exact analytic number theory gate. It is intentionally narrower than the full Erdős–Graham problem.

11 Dependency graph

The conditional architecture is:

PHNC $\Rightarrow$ finite smoothing $\Rightarrow$ Kummer collision control $\Rightarrow$ rank-two large sieve $\Rightarrow$ BVK $\Rightarrow$ NEG-203.

The first arrow is finite/probabilistic, the middle arrows are analytic, and the final arrow is sieve-theoretic. This decomposition is part of the claim: each dependency is visible and separately auditable.

12 Best-safe claim

The best-safe claim of this paper is:

A uniform fixed-radical Kummer non-concentration theorem in the above range would imply the rank-two Kummer distribution input required by the conditional Erdős–Graham #203 route, subject to the explicitly stated smoothing and sieve bridges.

This is not an unconditional theorem. It is a precise conditional research target.

13 Claim-separation table

Claim	Status	Note
Kummer independence	elementary	valuation proof
Low-range Chebotarev input	cited theorem needed	range must be checked
High- ℓ PHNC	open	main seam
PHNC $\Rightarrow$ KRWS	conditional bridge	finite lemma needed
Kummer-BV $\Rightarrow$ NEG-203	conditional bridge	sieve formalization needed
Full #203	not claimed	depends on all above

14 Conclusion

This paper identifies a precise analytic target. It does not claim that the target is currently proved in the high- ℓ seam. Its value is to make the route to #203 exact.

References

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